



# **NORTHERN UNIVERSITY**

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## **OF BUSINESS & TECHNOLOGY KHULNA**

**Department of Computer Science and Engineering**

**Project Title: AI-Based Smart Quiz System.**

Course Code: CSE 4204

Course Title: Mobile Computing Lab

**Submitted By:**

**Team Name** : CSE4204-8A-T03

**Section** : 8A

**GitHub Repository:** <https://github.com/sudipto224/AI-based-Smart-Quiz-System>

**Submitted to:**

Md. Riaz Mahmud

Designation: Assistant Professor

Department of CSE

**Date of Submission: 22/06/2026**

## **TEAM INFORMATION**

|                    |                                                                                                                                 |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Official Team Name | CSE4204-8A-T03                                                                                                                  |
| Section            | 8A                                                                                                                              |
| Project Title      | AI-Based Smart Quiz System                                                                                                      |
| GitHub Repository  | <a href="https://github.com/sudipto224/AI-based-Smart-Quiz-System">https://github.com/sudipto224/AI-based-Smart-Quiz-System</a> |

### **Team Information:**

| <b>Role</b>             | <b>Full Name</b>      | <b>Student ID</b> |
|-------------------------|-----------------------|-------------------|
| Team Leader + Backend   | Sudipto Das           | 11220320830       |
| Frontend                | Chaon Das             | 11210320629       |
| Artificial Intelligence | Rakibur Rahman Mishan | 11220320829       |
| Database                | Aditi Roy             | 11220320846       |

## **Project Information**

**Project Overview :** The AI-Based Smart Quiz System is a web-based online assessment platform built with Laravel 12, MySQL, and vanilla JavaScript. It is designed for academic institutions to conduct efficient, secure, and engaging quizzes. The system supports two user roles: Teacher and Student.

Teachers can create courses, manually add multiple-choice questions, and set a configurable timer per question. For a designated course, teachers can use an AI feature that generates complete MCQs automatically. By simply entering a topic, the system calls the Google Gemini API and returns a question, four options, the correct answer, and an explanation. The form is then auto-filled for the teacher to review and save.

Students can take timed quizzes for any available course. A countdown timer is shown for each question, and the quiz auto-submits if time runs out. After submission, students instantly see their score, the correct answers, and explanations for the questions they got wrong.

A unique AI-powered cheating detection system is active during quizzes for the designated course. Using JavaScript's `visibilitychange` event, the system counts how many times a student switches away from the quiz tab. It also calculates the average time spent per question. If the student switches tabs at least once or averages less than five seconds per question, the attempt is marked as suspicious. Teachers can review all flagged attempts in a dedicated panel.

The system also includes a leaderboard that ranks students by highest score and shortest completion time. Suspicious attempts are excluded from the leaderboard to ensure fairness.

Technology used includes HTML5, custom CSS, and vanilla JavaScript for the frontend; Laravel 12 with PHP 8.2 for the backend; MySQL 8.0 for the database; and Google Gemini API for AI question generation. Diagrams are created with draw.io and [dbdiagram.io](https://dbdiagram.io), and the project is managed with Git and GitHub.

This project is developed as part of the CSE4204 course at Northern University of Business and Technology, Khulna. It covers the complete quiz lifecycle, two AI features, and a

responsive user interface. Future enhancements such as real-time face proctoring, progressive web app support, and teacher analytics dashboards are not included in the current scope.

## **Objectives :**

The primary objectives of this project are as follows:

- **Automate Question Generation:** To reduce teacher workload by integrating Google Gemini API, which generates complete multiple-choice questions (including four options, correct answer, and explanation) from a simple topic input.
- **Ensure Academic Integrity:** To implement a robust cheating detection system using JavaScript's `visibilitychange` event and average time-per-question calculation. The system flags suspicious attempts (tab switching or average time below five seconds) and stores them with an `is_suspicious` flag for teacher review.
- **Provide Instant Feedback:** To display immediate quiz results with detailed explanations for each question, showing students their answers, the correct answers, and explanations.
- **Promote Engagement and Fairness:** To include a course-wise leaderboard that ranks students by highest score and shortest completion time, while excluding suspicious attempts.
- **Build a Scalable, Modular Platform:** To develop a maintainable web application using Laravel 12, MySQL, and vanilla JavaScript, following MVC architecture.

# 1. Wireframes or UI Designs

Landing/Home Page :

AI-Based Smart Quiz System

HomeAboutContact Us

LoginRegister

## AI-Powered Smart Quiz Platform

Generate questions with AI, detect cheating, and track performance in real-time.

Get StartedLearn More

### Key Features

AI Question Generation

Cheating Detection

Timed Quizzes

Instant Results

Leaderboard

Suspicious Review

### Our Team

Sudipto Das

Team Leader + Backend

Chaon Das

Frontend

Rakibur Rahman  
Mishan

Artificial Intelligence

Aditi Roy

Database

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## Login to Your Account

Email Address

Password

Login as:

☒ Teacher

☐ Student

[Forgot Password?](#)

Login

Don't have an account? Register Here [Register Here](#)

## Create Your Account

**Full Name**

**Email Address**

**Password**

**Confirm Password**

**Register as**

**Register**

Already have an account? [Login Here](#)

## Welcome, Student Name!

Select a course below to start your quiz.

### Available Courses

**Mobile Computing Lab**

10 Questions | 30 sec per question

[Start Quiz](#)

**Introduction to Computers**

8 Questions | 20 sec per question

[Start Quiz](#)

**Data Structures**

12 Questions | 25 sec per question

[Start Quiz](#)

### Leaderboard (Top 3)

[View All](#)

|              |                          |
|--------------|--------------------------|
| 1. Student A | Score: 9/10   Time: 120s |
| 2. Student B | Score: 8/10   Time: 135s |
| 3. Student C | Score: 7/10   Time: 150s |

Welcome, Teacher Name!

Manage your courses, questions, and monitor suspicious attempts.

Manage Courses  
Create, edit, delete courses

Manage Questions  
Add, edit, delete MCQs

AI Generator  
Generate questions with AI

Suspicious Attempts (Recent)

View All

| Student   Score   Tab Switches   Avg Time   Date |  |  |  |  |
|--------------------------------------------------|--|--|--|--|
| Rahim Uddin   6/10   2   4.2s   2026-06-20       |  |  |  |  |
| Sumaiya Akter   8/10   0   3.1s   2026-06-19     |  |  |  |  |
| Kamal Hossain   5/10   1   4.8s   2026-06-18     |  |  |  |  |

## My Profile

Full Name

Sudipto Das

Email Address

chaon@gmail.com

Role

Student

Edit Profile

Change Password

Mobile Computing Lab | Total Questions: 10 | 5:00 remaining

**Q1. What is the primary function of a Foreign Agent in Mobile IP?**

- |                                         |                                   |
|-----------------------------------------|-----------------------------------|
| A. Maintains a home address             | C. Encrypts all data transmission |
| B. Forwards packets to current location | D. Authenticates user password    |

**Q2. What is the brain of a computer?**

- |        |              |
|--------|--------------|
| A. CPU | C. Hard Disk |
| B. RAM | D. Monitor   |

**Q3. Which of the following is a volatile memory?**

- |        |              |
|--------|--------------|
| A. ROM | C. Hard Disk |
| B. RAM | D. SSD       |

**Q4. What is the difference between Stack and Queue?**

- |                                     |                    |
|-------------------------------------|--------------------|
| A. Stack uses LIFO, Queue uses FIFO | C. Both use LIFO   |
| B. Both use FIFO                    | D. Stack is faster |

... and 6 more questions (scrolling required)

Submit Quiz

Hidden fields: tab\_switch\_count, quiz\_start\_time

Your Score

6 / 10

☐ This attempt will be reviewed by the teacher.

[View Leaderboard](#)

**Q1. What is the primary function of a Foreign Agent in Mobile IP?**

Your answer: A | Correct answer: B | Explanation: The Foreign Agent acts as a router on the visited network.

**Q2. What is the brain of a computer?**

Your answer: A | Correct answer: A | Explanation: CPU executes instructions and is considered the brain of the computer.

**Q3. Which of the following is a volatile memory?**

Your answer: C | Correct answer: B | Explanation: RAM is volatile memory – it loses data when power is turned off.

**Q4. What is the difference between Stack and Queue?**

Your answer: A | Correct answer: A | Explanation: Stack uses LIFO (Last In First Out) and Queue uses FIFO (First In First Out).

... and 6 more questions (scroll to view all)

[Back to Dashboard](#)

## AI Question Generator

Enter Topic

e.g., Mobile IP, Handoff, Ad Hoc Networks

Generate with AI

### Generated Question Preview

Question

What is the primary function of a Foreign Agent in Mobile IP?

Option A

Maintains a home address for the mobile node

Option B

Forwards packets to the mobile node's current location

Correct Answer

**B. Forwards packets to the mobile node's current location**

Explanation

The Foreign Agent acts as a router on the visited network that delivers packets to the mobile node.

Save Question

## Suspicious Attempts Report

| Student Name   Score   Tab Switches   Avg Time   Attempted At |
|---------------------------------------------------------------|
| Rahim Uddin   6/10   2   4.2 sec   2026-06-20 10:30           |
| Sumaiya Akter   8/10   0   3.1 sec   2026-06-19 14:15         |
| Kamal Hossain   5/10   1   4.8 sec   2026-06-18 09:45         |
| Nadia Rahman   7/10   3   2.9 sec   2026-06-17 11:20          |
| Shahin Alam   9/10   0   4.5 sec   2026-06-16 16:00           |

Showing 1-5 of 12 | < Previous | Next >

## AI Interaction Hub

Question Generator

AI Chatbot

Coming Soon

### Generate MCQs with AI

Enter Topic

e.g., Mobile IP, Handoff, Ad Hoc Networks

Generate

### Generated Question Preview

What is the primary function of a Foreign Agent in Mobile IP?

- A. Maintains a home address for the mobile node
- B. Forwards packets to the mobile node's current location
- C. Encrypts all data transmission
- D. Authenticates the user's password

**Correct Answer: B. Forwards packets to the mobile node's current location**

Explanation: The Foreign Agent acts as a router on the visited network that delivers packets to the mobile node.

Save to Database

### AI Chatbot Support

Ask questions, get explanations, and clarify doubts with AI.

Coming Soon

Quiz Admin

Dashboard

Courses

Questions

AI Generator

Suspicious

Profile

Logout

Admin Dashboard

Admin Name

Courses12

Questions124

Suspicious18

AI Generated45

Quick Actions

Add Course

Add Question

AI Generate

View Suspicious

Recent Courses

Mobile Computing Lab | Questions: 10 | Timer: 30s

Introduction to Computers | Questions: 8 | Timer: 20s

Data Structures | Questions: 12 | Timer: 25s

View All Courses

Recent Suspicious Attempts

Rahim Uddin | Score: 6/10 | Tab Switches: 2 | Avg: 4.2s

Sumaiya Akter | Score: 8/10 | Tab Switches: 0 | Avg: 3.1s

Kamal Hossain | Score: 5/10 | Tab Switches: 1 | Avg: 4.8s

View All Suspicious

AI Usage Stats

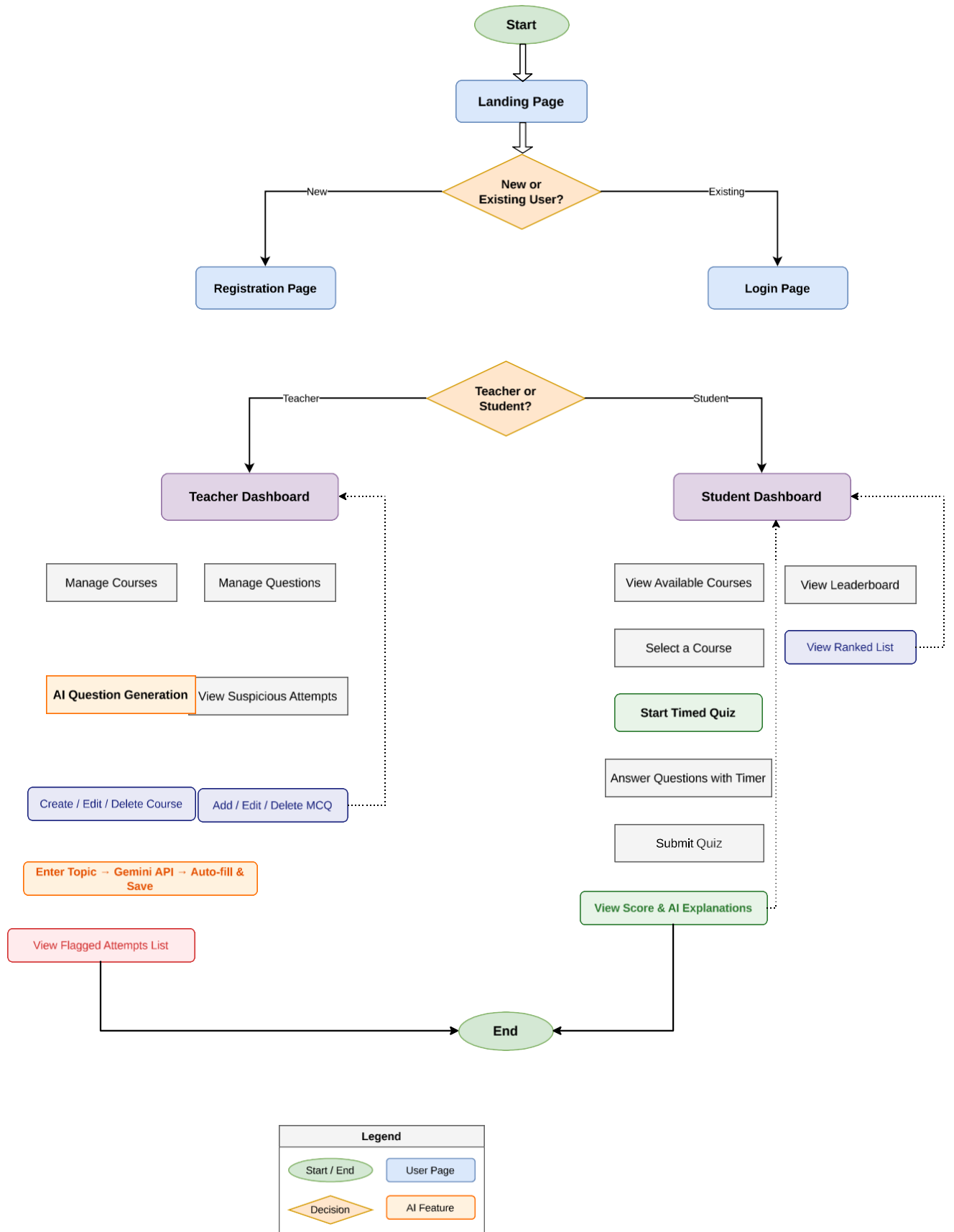
Total Questions Generated: 45

API Calls (This Month): 120

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## 2. User Flow Diagram

### User Flow Diagram - AI-Based Smart Quiz System



### **3. Development Roadmap – AI-Based Smart Quiz System**

This roadmap outlines the structured development plan for the AI-Based Smart Quiz System from Week 06 to Week 11 of the semester. The roadmap adopts a phased approach, whereby each week targets a distinct layer of the system—progressing from backend architecture and database engineering through frontend implementation, AI integration, feature completion, quality assurance, and, finally, deployment. Responsibilities are distributed among team members in alignment with their designated roles, ensuring parallel workstreams where feasible and clear accountability at each stage. The schedule has been designed to accommodate iterative review and testing throughout the development cycle, thereby minimising integration risk ahead of the final submission.

#### **Phased Development Schedule**

| <b>Week</b> | <b>Module / Task</b>                           | <b>Team Member(s)<br/>Responsible</b> | <b>Key Deliverables</b>                                                                                                                                          |
|-------------|------------------------------------------------|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 06          | Backend Development &<br>Database Architecture | Sudipto Das, Aditi Roy                |                                                                                                                                                                  |
| 07          | Frontend Implementation                        | Chaon Das                             | Responsive Blade/Vue layouts for Teacher and Student dashboards; course listing and management views; MCQ creation form; quiz-taking interface with per-question |

| Week | Module / Task                           | Team Member(s)<br>Responsible         | Key Deliverables                                                                                                                                                                                                                                                                   |
|------|-----------------------------------------|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |                                         |                                       | countdown timer; result and explanation pages; leaderboard display.                                                                                                                                                                                                                |
| 08   | AI Service Integration – Gemini API     | Rakibur Rahman<br>Mishan, Sudipto Das | Google Gemini API client integration; automatic MCQ generation endpoint for Mobile Computing Lab; AI-generated question review and storage workflow; error handling and rate-limit management for the Gemini service.                                                              |
| 09   | Feature Completion & Cheating Detection | Sudipto Das,<br>Chaon Das             | Per-question timer logic (configurable per course); tab-switch detection via JavaScript visibilitychange event; is_suspicious flag logic (average time per question < 5 seconds); Teacher dashboard alerts for flagged attempts; end-to-end feature integration and smoke testing. |
| 10   | Testing, Debugging & Quality Assurance  | All Members                           | Unit and feature tests (PHPUnit); frontend usability testing; AI response validation; cheating-detection edge-case testing; bug triage and resolution; performance                                                                                                                 |

| Week | Module / Task                 | Team Member(s) Responsible | Key Deliverables                                                                                                                                                                                                      |
|------|-------------------------------|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |                               |                            | review; code refactoring and documentation.                                                                                                                                                                           |
| 11   | Deployment & Final Submission | All Members                | Production environment setup; database seeding with sample data; final integration testing on live server; project report finalisation; presentation preparation; submission of source code, documentation, and demo. |

#### Detailed Module-Wise Responsibilities

| Module           | Scope of Work                                                                                                                                                                                                                                                            | Assigned Team Member(s) |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| Backend Logic    | Laravel 12 application architecture; MVC implementation; role-based authentication and middleware; RESTful API endpoints for quiz management, attempt recording, and leaderboard aggregation; cheating-detection service logic; timer configuration per course.          | Sudipto Das             |
| Frontend & UI/UX | Blade templating and/or Vue.js component design; responsive layouts for Teacher and Student portals; interactive quiz-taking interface with live per-question countdown; result and explanation views; leaderboard table; accessibility and cross-browser compatibility. | Chaon Das               |

| Module                  | Scope of Work                                                                                                                                                                                                                                             | Assigned Team Member(s) |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| Database Engineering    | Relational schema design for users, courses, questions, options, quiz attempts, and leaderboard tables; Eloquent migrations and seeders; query optimisation; indexing strategy; data integrity constraints; Entity-Relationship (ER) diagram maintenance. | Aditi Roy               |
| Artificial Intelligence | Google Gemini API integration for automated MCQ generation targeting the Mobile Computing Lab course; prompt engineering for question quality; parsing and storing AI-generated content; fallback and validation logic for API responses.                 | Rakibur Rahman Mishan   |
| Testing & QA            | Unit tests for backend services and controllers (PHPUnit); feature tests for quiz flow and cheating detection; frontend usability and regression testing; AI output validation; bug reporting and resolution tracking; final pre-deployment verification. | All Members             |

**Milestone Summary**

| Milestone                   | Tentative Completion | Status                                        |
|-----------------------------|----------------------|-----------------------------------------------|
| UI/UX Design Finalisation   | Week 05              | <input checked="" type="checkbox"/> Completed |
| Backend & Database Setup    | Week 06              | <input type="checkbox"/> In Progress          |
| Frontend Implementation     | Week 07              | <input type="checkbox"/> Pending              |
| AI Integration (Gemini API) | Week 08              | <input type="checkbox"/> Pending              |

| Milestone                               | Tentative Completion | Status    |
|-----------------------------------------|----------------------|-----------|
| Feature Completion & Cheating Detection | Week 09              | ☐ Pending |
| Testing & Debugging                     | Week 10              | ☐ Pending |
| Deployment & Presentation               | Week 11              | ☐ Pending |

### Deliverables Overview

The following summarises the primary deliverable(s) expected at the conclusion of each development week:

- Week 06 – Backend & Database: Fully configured Laravel 12 application with role-based authentication, complete database schema, Eloquent models, migrations, and core API routes.
- Week 07 – Frontend: Responsive Teacher and Student dashboards, quiz-taking interface with per-question countdown timer, results page, and leaderboard view.
- Week 08 – AI Integration: Operational Google Gemini API integration enabling automatic MCQ generation for the Mobile Computing Lab course, with validated question storage.
- Week 09 – Feature Completion: Fully functional cheating-detection system (tab-switch tracking, is\_suspicious flag), configurable per-question timer, and integrated end-to-end application flow.
- Week 10 – Testing & QA: Comprehensive test suite (unit and feature), resolved bug backlog, performance review report, and refactored, documented codebase.
- Week 11 – Deployment & Submission: Live deployment on production server, seeded demo data, final project report, presentation slides, and complete submission package (source code, documentation, and demo).

#### **4. Team Task Distribution**

A clearly defined task distribution is fundamental to the successful delivery of any collaborative software project. For the AI-Based Smart Quiz System, responsibilities have been assigned in direct alignment with each team member's declared expertise and role, as established in the Software Requirements Specification (SRS). This structured approach ensures accountability, minimises duplication of effort, and enables parallel workstreams across the development lifecycle. The distribution outlined in this section covers primary role assignments, a task-level responsibility matrix, a week-by-week breakdown of lead responsibilities, and individual contribution summaries for all four members of Team CSE4204-8A-T03.

##### **Primary Role Assignments**

| Member Name | Primary Role                    | Key Responsibilities                                                                                                                                                                                                                                                                                                                            |
|-------------|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sudipto Das | Team Leader & Backend Developer | Overall project coordination and task management; Laravel 12 application architecture; implementation of controllers, routes, and middleware; role-based authentication logic using Laravel Breeze; backend logic for cheating detection (is_suspicious flag); API endpoint design and integration; code review and final deployment oversight. |
| Chaon Das   | Frontend Developer              | Design and implementation of all Blade view templates; custom CSS styling for Teacher and Student portals; Vanilla JavaScript development for interactive features including the per-question countdown timer and tab-switch detection (visibilitychange event); UI/UX wireframing and user flow design; cross-browser compatibility and        |

| Member Name              | Primary Role                 | Key Responsibilities                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          |                              | responsiveness.                                                                                                                                                                                                                                                                                                                                                        |
| Rakibur Rahman<br>Mishan | AI Integration<br>Specialist | Integration of the Google Gemini API (gemini-2.0-flash) into the Laravel backend; prompt engineering for accurate and contextually relevant MCQ generation for the Mobile Computing Lab course; parsing and validating JSON responses from the Gemini API; error handling and rate-limit management for the AI service; testing AI-generated question quality.         |
| Aditi Roy                | Database<br>Manager          | Design of the relational MySQL 8.0 schema covering users, courses, questions, options, quiz attempts, and leaderboard tables; creation and maintenance of Laravel Eloquent migrations and seeders; definition of model relationships (one-to-many, many-to-many); query optimisation and indexing strategy; Entity-Relationship (ER) diagram design and documentation. |

### Task Distribution Matrix

| Task / Module                                                           | Lead Member | Supporting Member(s)                 |
|-------------------------------------------------------------------------|-------------|--------------------------------------|
| Backend Development<br>(Laravel Controllers,<br>Routes, Authentication) | Sudipto Das | Rakibur Rahman Mishan (AI endpoints) |
| Frontend Development<br>(Blade Views, CSS,<br>JavaScript)               | Chaon Das   | Sudipto Das (routing integration)    |
| Database Design &                                                       | Aditi Roy   | Sudipto Das (Eloquent usage)         |

| Task / Module                                            | Lead Member                  | Supporting Member(s)                 |
|----------------------------------------------------------|------------------------------|--------------------------------------|
| Management (Schema, Migrations, Models)                  |                              |                                      |
| AI Integration (Gemini API, Prompt Engineering)          | Rakibur Rahman<br>Mishan     | Sudipto Das (backend service layer)  |
| Cheating Detection (JS visibilitychange + Backend Logic) | Sudipto Das                  | Chaon Das (client-side JS)           |
| UI/UX Design (Wireframes, User Flow, Diagrams)           | Chaon Das                    | All Members (review & feedback)      |
| Documentation (SRS, System Design, API Design, Reports)  | Sudipto Das                  | All Members (section contributions)  |
| Testing & Quality Assurance (Unit Testing, Bug Fixing)   | All Members                  | Sudipto Das (coordination)           |
| Deployment & DevOps (Server Setup, Live Deployment)      | Sudipto Das                  | All Members (environment validation) |
| Project Management & Coordination                        | Sudipto Das<br>(Team Leader) | All Members                          |

### Responsibility Breakdown by Week

| Week | Lead Member(s)         | Key Tasks                                                                                                                                                      |
|------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 06   | Sudipto Das, Aditi Roy | Laravel 12 project scaffolding and configuration; role-based authentication implementation via Laravel Breeze; complete MySQL database schema design; Eloquent |

| Week | Lead Member(s)                        | Key Tasks                                                                                                                                                                                                                                                                                                                                    |
|------|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |                                       | model creation and migration setup; API route definitions for core modules; initial repository structure and branching strategy on GitHub.                                                                                                                                                                                                   |
| 07   | Chaon Das                             | Development of all Blade view templates for Teacher and Student portals; custom CSS layout and styling; JavaScript implementation of the per-question countdown timer; course listing and MCQ management UI; quiz-taking interface; results and explanation views; leaderboard display page.                                                 |
| 08   | Rakibur Rahman<br>Mishan, Sudipto Das | Google Gemini API client integration within the Laravel service layer; prompt engineering and refinement for Mobile Computing Lab MCQ generation; JSON response parsing and database storage workflow; error handling and fallback logic for API failures; end-to-end testing of AI question generation pipeline.                            |
| 09   | Sudipto Das, Chaon Das                | Client-side tab-switch detection using the JavaScript visibilitychange event; server-side cheating evaluation logic (is_suspicious flag for average time < 5 seconds per question); Teacher dashboard alerts for suspicious attempts; configurable per-question timer logic; full integration of all application features and smoke testing. |
| 10   | All Members                           | Comprehensive unit and feature testing (PHPUnit); frontend usability and regression testing; AI response validation and edge-case handling; cheating-detection boundary testing; bug triage and resolution; performance review; code refactoring; technical documentation finalisation; deployment platform evaluation and selection.        |
| 11   | All Members                           | Production environment setup and configuration on                                                                                                                                                                                                                                                                                            |

| Week | Lead Member(s)               | Key Tasks                                                                                                                                                                                                                                                                             |
|------|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      | (Coordinated by Sudipto Das) | selected hosting platform; database seeding with representative sample data; final integration testing on live server; project report compilation and proofreading; preparation of demonstration slides; formal submission of source code, documentation package, and live demo link. |

### Summary of Individual Contributions

**Sudipto Das (Team Leader & Backend Developer):** Sudipto bears overarching responsibility for the project’s technical direction and team coordination. As the primary backend developer, he architects and implements the Laravel 12 application framework, including all controllers, routes, middleware, and role-based authentication. He leads the cheating-detection backend logic, oversees API design, and coordinates the final deployment. His leadership role also encompasses task scheduling, progress monitoring, and quality assurance oversight across all team activities.

**Chaon Das (Frontend Developer):** Chaon is responsible for the entirety of the user-facing interface, translating system requirements into functional, responsive Blade templates styled with custom CSS. He implements all client-side JavaScript features, most notably the per-question countdown timer and the tab-switch detection mechanism using the browser’s visibilitychange API. Chaon also leads the UI/UX design process, producing wireframes and user-flow diagrams that guide the visual and interaction design of both the Teacher and Student portals.

**Rakibur Rahman Mishan (AI Integration Specialist):** Rakibur’s principal contribution lies in the integration of the Google Gemini API into the system’s backend. He is responsible for designing and refining the prompts used to generate high-quality, curriculum-relevant MCQs for the Mobile Computing Lab course, parsing the structured JSON responses returned by the API, and storing validated questions in the database. He also implements appropriate error handling and rate-limit management to ensure the AI service operates reliably within the application.

**Aditi Roy (Database Manager):** Aditi owns the complete database layer of the system. She designs the relational MySQL 8.0 schema to accommodate all entities—including users, courses, questions, answer options, quiz attempts, and leaderboard data—and implements the corresponding Eloquent migrations, seeders, and model relationships. Her work ensures the integrity, consistency, and efficiency of all data operations within the application. She also maintains the ER diagram as a living design document throughout the development cycle.

### **Collaboration & Communication**

The team will maintain structured communication and collaboration throughout the project using a combination of synchronous and asynchronous channels. Weekly progress meetings will be held to review completed work, address blockers, and align on forthcoming tasks. Day-to-day communication and file sharing will be conducted via a dedicated WhatsApp group and/or Discord server, ensuring all members remain informed of developments in real time. All source code will be managed through a shared GitHub repository, with a branching strategy that enables members to work concurrently on separate features without risk of conflict. Pull requests will be subject to peer review prior to merging into the main branch, maintaining code quality and shared awareness of all contributions. Task status and issue tracking will be monitored collaboratively, with Sudipto Das coordinating weekly check-ins to ensure the project remains on schedule.

### **Conclusion**

Team CSE4204-8A-T03 has established a comprehensive and equitable task distribution that leverages each member's individual strengths whilst ensuring collective accountability for the project's success. Sudipto Das provides both technical leadership and backend expertise; Chaon Das drives the frontend experience and UI/UX design; Rakibur Rahman Mishan delivers the AI integration capability central to the system's intelligence; and Aditi Roy underpins the entire application with a robust and well-structured database design. With clearly delineated responsibilities, a collaborative communication framework, and a structured weekly schedule, the team is well-positioned to deliver the AI-Based Smart Quiz System to a high standard within the allotted semester timeline.

## 5. Final Technology Stack

The technology stack for the AI-Based Smart Quiz System has been finalised based on the project requirements documented in the Software Requirements Specification (SRS) and the team's collective technical proficiency. Each technology choice reflects a deliberate decision informed by factors such as compatibility with the academic environment, cost-effectiveness, ease of integration, and suitability for the specific functional requirements of the system. This section presents a comprehensive overview of the selected technologies, the rationale underpinning each choice, and the planned deployment strategy.

### Technology Stack Overview

| Layer    | Technology                                     | Version                        | Purpose / Reason                                                                                                                                                                                                                                                      |
|----------|------------------------------------------------|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Frontend | HTML5,<br>Custom CSS,<br>Vanilla<br>JavaScript | HTML5<br>/ CSS3 /<br>ES6+      | Provides the user interface layer for both Teacher and Student portals. Vanilla JavaScript handles dynamic interactions such as the per-question countdown timer and tab-switch detection, without the overhead of a frontend framework.                              |
| Backend  | Laravel 12<br>(PHP)                            | Laravel<br>12 /<br>PHP<br>8.2+ | Serves as the core application framework, managing routing, MVC architecture, business logic, session handling, and API communication with the Gemini service. Laravel's expressive syntax and rich ecosystem accelerate development whilst maintaining code quality. |
| Database | MySQL 8.0                                      | MySQL<br>8.0                   | Stores all persistent data including users, courses, MCQs, quiz attempts, and leaderboard records. MySQL 8.0 offers mature support for relational integrity, indexing, and JSON columns, and integrates seamlessly with Laravel's                                     |

| Layer              | Technology                                    | Version                      | Purpose / Reason                                                                                                                                                                                                                                                                             |
|--------------------|-----------------------------------------------|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    |                                               |                              | Eloquent ORM.                                                                                                                                                                                                                                                                                |
| Authentication     | Laravel Breeze (session-based)                | Laravel Breeze (latest)      | Provides a lightweight, pre-built authentication scaffold covering registration, login, and password management. Session-based authentication is well-suited to a server-rendered web application and simplifies role-based access control for Teacher and Student roles.                    |
| AI Integration     | Google Gemini API (gemini-2.0-flash)          | Gemini 2.0 Flash (free tier) | Powers the automatic MCQ generation feature for the Mobile Computing Lab course. The gemini-2.0-flash model offers a strong balance between response speed and output quality, and is accessible at no cost under the free-tier quota, making it ideal for an academic project.              |
| Cheating Detection | JavaScript visibilitychange + Laravel backend | ES6+ / Laravel 12            | Detects tab-switching behaviour on the client side using the browser's visibilitychange event. The Laravel backend evaluates average time per question and sets the is_suspicious flag when the average falls below five seconds per question, enabling Teachers to review flagged attempts. |
| Version Control    | Git & GitHub                                  | Git 2.x                      | Provides distributed source-code management, branch-based collaborative development, and a complete audit trail of all changes. GitHub serves as the shared remote repository and facilitates code review via pull requests.                                                                 |

| Layer                    | Technology                                   | Version                      | Purpose / Reason                                                                                                                                                                                                                                                 |
|--------------------------|----------------------------------------------|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Diagrams & Documentation | Draw.io,<br>Mermaid.js,<br>dbdiagram.io      | Latest<br>(web-based)        | Draw.io supports system architecture and flowchart diagrams; Mermaid.js enables version-controlled, text-based diagrams embedded in Markdown; dbdiagram.io is used for Entity-Relationship (ER) diagram creation and database schema visualisation.              |
| Deployment               | To be decided<br>(Vercel / Render / Railway) | TBD –<br>Decision in Week 10 | The production hosting platform will be finalised in Week 10 following a comparative evaluation of Vercel, Render, and Railway. All three options support PHP/Laravel deployments and offer free or low-cost tiers suitable for academic demonstration purposes. |

### Why This Stack?

The following points articulate the key rationale behind the technology selections made for this project:

- **HTML5, Custom CSS, and Vanilla JavaScript:** A framework-free frontend approach was deliberately selected to reduce build-tool complexity and dependency overhead. Given the scope of the project and the team's familiarity with standard web technologies, Vanilla JavaScript provides sufficient capability for all required client-side features, including the countdown timer and cheating-detection logic, without the learning curve associated with frameworks such as React or Vue.js.
- **Laravel 12:** Laravel was selected as the backend framework owing to its mature MVC architecture, expressive Eloquent ORM, built-in routing and middleware support, and its seamless integration with MySQL and Breeze authentication. Laravel 12's compatibility with PHP 8.2+ ensures access to modern language features whilst benefiting from a well-established ecosystem of packages and community support.

- **MySQL 8.0:** As a proven relational database management system, MySQL 8.0 offers robust support for the structured data models required by this application, including relational integrity between users, courses, questions, and quiz attempts. Its wide adoption, strong Laravel integration via Eloquent, and availability on most hosting platforms made it the natural choice for this project.
- **Laravel Breeze:** Breeze provides a minimal, clean authentication scaffold that is straightforward to extend with custom role-based access control logic. Its session-based approach aligns well with the server-rendered nature of the application and avoids the additional complexity of token-based or OAuth authentication systems, which are unnecessary for the current scope.
- **Google Gemini API (gemini-2.0-flash):** The Gemini API was chosen for its accessibility under a free-tier quota, which is appropriate for an academic project with no allocated budget for AI services. The gemini-2.0-flash model delivers fast, high-quality text generation, making it well-suited for producing accurate and contextually relevant MCQs for the Mobile Computing Lab course. Its straightforward REST interface integrates cleanly with Laravel's HTTP client.
- **JavaScript visibilitychange Event + Laravel Backend:** The browser's native visibilitychange API provides a reliable, lightweight mechanism for detecting tab-switching behaviour without requiring any third-party library. Combining this with server-side validation logic in Laravel ensures that cheating detection is both tamper-resistant and auditable, as the is\_suspicious flag is set and persisted in the database rather than relying solely on client-side state.
- **Git & GitHub:** Git is the industry-standard version control system and provides the team with robust branching, merging, and rollback capabilities essential for collaborative development. GitHub serves as the centralised remote repository and enables code review, issue tracking, and transparent contribution history across all team members.
- **Draw.io, Mermaid.js, and dbdiagram.io:** These free, web-based diagramming tools collectively cover all documentation needs for the project. Draw.io is used for high-level architecture and process flowcharts; Mermaid.js enables diagrams to be maintained as code within the project repository; and dbdiagram.io provides a

dedicated, intuitive interface for designing and communicating the database schema to all stakeholders.

## Deployment Strategy

The deployment platform for the AI-Based Smart Quiz System has not yet been finalised and will be determined during Week 10 of the project schedule, following a structured comparative evaluation by the team. Three platforms have been shortlisted for consideration:

- **Vercel:** Primarily recognised as a frontend hosting platform, Vercel also supports serverless functions and can host Laravel applications via community adapters. It offers a generous free tier and excellent continuous deployment integration with GitHub.
- **Render:** A cloud platform that natively supports PHP web services, background workers, and managed databases. Render's free tier accommodates small Laravel deployments and provides automatic deployment from GitHub repositories, making it a practical option for academic projects.
- **Railway:** A developer-focused deployment platform that supports Laravel out of the box, with integrated MySQL provisioning. Railway's straightforward configuration and GitHub-linked deployments make it particularly accessible for teams without dedicated DevOps expertise.

The final platform selection will be based on ease of Laravel and MySQL configuration, free-tier resource limits, deployment reliability, and any constraints identified during the testing phase in Week 10. Once the platform is confirmed, environment variables, database connection settings, and the Gemini API key will be configured securely in the chosen hosting environment prior to the final demonstration.

## Conclusion

The technology stack for the AI-Based Smart Quiz System is well-defined, purposefully selected, and ready for implementation. Each component has been chosen to align with the functional and non-functional requirements of the system, the team's technical capabilities, and the academic constraints of the project. The combination of Laravel 12, MySQL 8.0, Vanilla JavaScript, Laravel Breeze, and the Google Gemini API provides a coherent, cost-

effective, and maintainable architecture that supports all core features, including AI-driven question generation, timed quizzes, role-based access control, and cheating detection. With the deployment decision scheduled for Week 10, the stack will be fully resolved ahead of the final submission, ensuring a smooth and timely delivery of the completed system.